Thermally Reversible Beta-Sheet-like Ordering Controls Gelation in Elastin-like Polypeptides

Doga Ozgulbas,[†] Miaoqi Chu,[‡] Alex Lavens,[‡] Nicholas Marks,[†] Ming Du,[‡] Xiaobing Zuo,[‡] Michael Irvin,[†] Heng Ma,[†] Soenke Seifert,[‡] Byeongdu Lee,[‡] Rafael Vescovi,[†] Ian Foster,[†] Victor Mateevitsi,[¶] Don Jensen Jr.,[‡] Chase Akins,[§] Arvind Ramanathan,[*,†] Gyorgy Babnigg,[*,§] and Qingteng Zhang[*,‡]

[†]*Data Science and Learning Division, Argonne National Laboratory, Lemont, Illinois 60439, United States*

[‡]*X-ray Science Division, Argonne National Laboratory, Lemont, Illinois 60439, United States*

[¶]*Argonne Leadership Computing Facility, Argonne National Laboratory, Lemont, Illinois 60439, United States*

[§]*Biosciences Division, Argonne National Laboratory, Lemont, Illinois 60439, United States*

E-mail: ramanathana@anl.gov; gbabnigg@anl.gov; qzhang234@anl.gov

Abstract

Amyloid-like β-sheet packing spans a wide range of stabilities, from the kinetically persistent cores of pathological fibrils to contacts that come apart again, and where a sequence falls in that range decides whether it can serve as a switchable material. Here, temperature-dependent small- and wide-angle X-ray scattering of the sequence-defined elastin-like polypeptide $(VPAVG)_{30}$ resolves a reversible transition between a dispersed low-temperature state and a hierarchical high-temperature assembly carrying β-sheet-like peaks at $Q = 1.38$ and 0.72 Å^{-1} that vanish on cooling. Chain length

decides whether the gel returns: $(VPAVG)_{30}$ recovers its low-temperature profile through repeated cycling while the ten-repeat construct does not. The reversible case therefore places amyloid-like local order in the role of a physical junction rather than a terminal aggregate.

Introduction

Amyloid-like β-sheet ordering occupies an unusual position in protein science and polymer materials. Pathological amyloid fibrils formed by proteins and peptides that share no sequence homology are associated with Alzheimer's disease,[1] Parkinson's disease,[2] type II diabetes,[3] prion diseases,[4] and systemic light-chain amyloidosis.[5] Many amyloid fibrils are kinetically persistent because extended hydrogen-bonded sheets are reinforced by tightly packed steric-zipper interfaces.[6] Yet amyloid-like order is not intrinsically pathological: functional amyloids store mammalian peptide hormones in pituitary secretory granules[7] and build the extracellular matrix of bacterial biofilms,[8] and related assemblies are increasingly used as mechanically robust biomaterials.[9] What differs across these systems is not whether β-sheet-like order forms but how long it lasts: hormone granules disassemble to release functional monomer, whereas steric-zipper cores are far more persistent. Amyloid-like packing therefore spans a wide range of stabilities rather than committing a sequence to one permanent structure,[10] and where a sequence falls in that range is a materials-design variable.

Reversible β-sheet-like contacts are attractive as candidate physical junctions because backbone hydrogen bonding can stabilize locally ordered assemblies, whereas hydrated or weakly packed interfaces can remain dynamic or reversible.[10–13] Elastin-like polypeptides (ELPs) are well suited to test this possibility because their sequence, molecular weight, and lower critical solution temperature (LCST) behavior can be programmed recombinantly.[14–18] Above the LCST, ELPs undergo hydrophobic desolvation and can form coacervates or hierarchical assemblies depending on sequence, concentration, and thermal history.[17–19] In particular, alanine-containing ELPs can form stiff, thermoresponsive gels through arrested

phase separation.[20,21] Although these studies establish the arrested state from time-averaged scattering and rheology, how the assembly evolves toward arrest remains elusive.

Results and Discussion

Figure 1 establishes the low- and high-temperature structures and their reversibility. Two aliquots from the same stock were measured at APS 12-ID-B by simultaneous SAXS/WAXS at 13.3 keV.[22–24] The "Measurement" aliquot had undergone more than 10 heating/cooling cycles; the "Reference" aliquot had stayed at 4 °C. At 10 °C the two overlap, and a Guinier fit to the Reference gives an apparent $R_g = 22.65 \pm 0.50$ Å (SI Appendix), a dispersed state with no mesoscale aggregates down to $Q = 0.0081$ Å^{-1}, which corresponds to structures below approximately 78 nm. At 30 °C, both aliquots reproducibly develop an approximate $I(Q) \propto Q^{-3.2}$ response at low Q (slopes -3.17 and -3.15 over $0.012 < Q < 0.040$ Å^{-1}). Both also develop the same pair of high-Q features, at $Q = 1.38$ and 0.72 Å^{-1} in the Reference and 1.36 and 0.74 Å^{-1} in the Measurement. The cycled aliquot alone makes the point: it still develops both peaks at 30 °C and shows neither at 10 °C. A more direct test at 8-ID-I took a single aliquot through seven consecutive 6–34 °C cycles, heated at 1 °C min^{-1} (SI Appendix). Across the seven cycles the 6 °C profile recovers to within a 9% relative standard deviation with no systematic drift, and at 33.8 °C the correlation function stays close to $1 + \beta$ in every cycle, the signature of an arrested state. The sample therefore assembled and arrested on every heating and returned to the same low-temperature state on every cooling (SI Appendix). Video S1 independently shows recovery of optical transparency after cooling.

The two reciprocal-space regimes describe a hierarchical assembly. Power-law small-angle scattering signifies density heterogeneity extending across the probed interval rather than a single resolved structural scale.[25] Related gel-forming ELPs exhibit disordered bicontinuous or fractal protein-rich networks produced by arrested phase separation.[20,21] The 4.5 and 8.8 Å spacings are compatible with strand-to-strand and sheet-to-sheet packing in cross-β-like

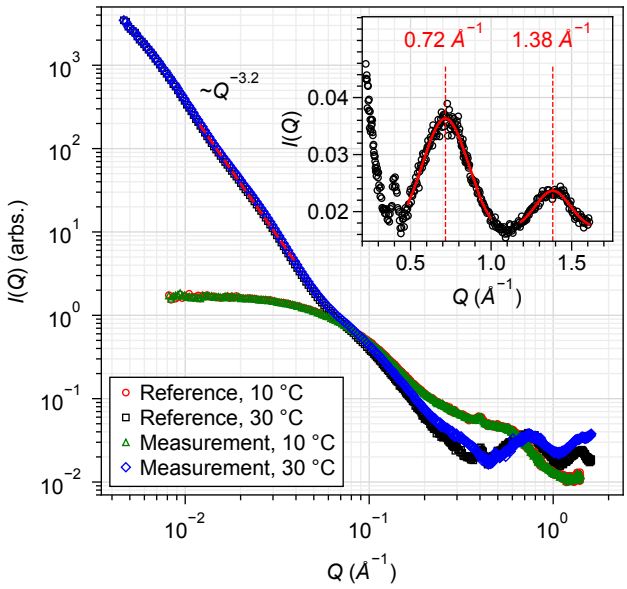


Figure 1: Temperature-dependent structure and reversibility of $(\mathrm{VPAVG})_{30}$. At 10 °C, the Measurement and Reference profiles overlap and the Reference data give an apparent $R_g = 22.65 \pm 0.50$ Å (SI Appendix). Heating produces an approximate $Q^{-3.2}$ low-Q response and peaks at $Q = 1.38$ and 0.72 Å^{-1}, corresponding to real-space spacings of approximately 4.5 and 8.8 Å, respectively. The dashed line is an unweighted power-law fit to the Reference profile at 30 °C over $0.012 < Q < 0.040$ Å^{-1}. Inset: the high-Q region of that same profile (open circles, same arbitrary intensity units as the main panel); red curves are independent Gaussian-plus-linear-background fits over $0.48 < Q < 1.00$ and $1.18 < Q < 1.70$ Å^{-1}, and the red dashed lines and labels mark the fitted peak centers. The recovery and reproducible re-formation of these signatures demonstrate a thermally reversible structural transition within experimental sensitivity.

assemblies, whose reflections are reported at approximately 4.8 Å along the fibril direction and between 8 and 11 Å across it.[6,26] Both features are broad and exhibit full widths at half maximum of 0.25 and 0.34 $Å^{-1}$ (correlation lengths $2\pi/\Delta Q$ of only about 25 and 19 Å), so the ordering extends over a few repeat spacings rather than forming extended cross-β crystallites. Their concurrent appearance with the low-Q response establishes coexistence of local ordered contacts and mesoscale density heterogeneity. The disappearance of the peaks on cooling identifies thermally labile β-sheet-like associations that are plausible, rather than uniquely established, physical junctions.

Materials and Methods

To be assembled from the pre-split Letter (tag `pre-split` in PubCode-2021-IDPP): the 12-ID-B paragraph, the 8-ID-I cell and beam description, and the robotic cell-assembly method.

Supporting Information

Recombinant synthesis and purification; sample environments and the automated cell assembly; SAXS/WAXS calibration and merging; Guinier analysis; chain-length ramps; seven-cycle thermal repeatability; irreversibility of the ten-repeat construct; videos of the reversible transition and of the robotic assembly (PDF, MP4).

Data and Code Availability

Reduced data and analysis code are at `https://github.com/qzhang234/PubCode-2021-IDPP-Reversibil`

References

(1) Masters, C. L.; Simms, G.; Weinman, N. A.; Multhaup, G.; McDonald, B. L.; Beyreuther, K. Amyloid Plaque Core Protein in Alzheimer Disease and Down Syndrome. *Proceedings of the National Academy of Sciences of the United States of America* **1985**, *82*, 4245–4249.

(2) Serpell, L. C.; Berriman, J.; Jakes, R.; Goedert, M.; Crowther, R. A. Fiber Diffraction of Synthetic α-Synuclein Filaments Shows Amyloid-Like Cross-β Conformation. *Proceedings of the National Academy of Sciences of the United States of America* **2000**, *97*, 4897–4902.

(3) Cooper, G. J. S.; Willis, A. C.; Clark, A.; Turner, R. C.; Sim, R. B.; Reid, K. B. M. Purification and Characterization of a Peptide from Amyloid-Rich Pancreases of Type 2 Diabetic Patients. *Proceedings of the National Academy of Sciences of the United States of America* **1987**, *84*, 8628–8632.

(4) Pan, K.-M.; Baldwin, M.; Nguyen, J.; Gasset, M.; Serban, A.; Groth, D.; Mehlhorn, I.; Huang, Z.; Fletterick, R. J.; Cohen, F. E.; Prusiner, S. B. Conversion of Alpha-Helices into Beta-Sheets Features in the Formation of the Scrapie Prion Proteins. *Proceedings of the National Academy of Sciences of the United States of America* **1993**, *90*, 10962–10966.

(5) Radamaker, L.; Lin, Y.-H.; Annamalai, K.; Huhn, S.; Hegenbart, U.; Schönland, S. O.; Fritz, G.; Schmidt, M.; Fändrich, M. Cryo-EM Structure of a Light Chain-Derived Amyloid Fibril from a Patient with Systemic AL Amyloidosis. *Nature Communications* **2019**, *10*, 1103.

(6) Sawaya, M. R.; Sambashivan, S.; Nelson, R.; Ivanova, M. I.; Sievers, S. A.; Apostol, M. I.; Thompson, M. J.; Balbirnie, M.; Wiltzius, J. J. W.; McFarlane, H. T.; Madsen, A. Ø.;

Riekel, C.; Eisenberg, D. Atomic Structures of Amyloid Cross-β Spines Reveal Varied Steric Zippers. *Nature* **2007**, *447*, 453–457.

(7) Maji, S. K.; Perrin, M. H.; Sawaya, M. R.; Jessberger, S.; Vadodaria, K.; Rissman, R. A.; Singru, P. S.; Nilsson, K. P. R.; Simon, R.; Schubert, D.; Eisenberg, D.; Rivier, J.; Sawchenko, P.; Vale, W.; Riek, R. Functional Amyloids as Natural Storage of Peptide Hormones in Pituitary Secretory Granules. *Science* **2009**, *325*, 328–332.

(8) Chapman, M. R.; Robinson, L. S.; Pinkner, J. S.; Roth, R.; Heuser, J.; Hammar, M.; Normark, S.; Hultgren, S. J. Role of Escherichia coli Curli Operons in Directing Amyloid Fiber Formation. *Science* **2002**, *295*, 851–855.

(9) Knowles, T. P. J.; Mezzenga, R. Amyloid Fibrils as Building Blocks for Natural and Artificial Functional Materials. *Advanced Materials* **2016**, *28*, 6546–6561.

(10) Pálmadóttir, T.; Getachew, J.; Ortigosa-Pascual, L.; Axell, E.; Wei, J.; Olsson, U.; Knowles, T. P. J.; Linse, S. On the Reversibility of Amyloid Fibril Formation. *Biophysics Reviews* **2025**, *6*, 011303.

(11) Kato, M. et al. Cell-free Formation of RNA Granules: Low Complexity Sequence Domains Form Dynamic Fibers within Hydrogels. *Cell* **2012**, *149*, 753–767.

(12) Hughes, M. P.; Sawaya, M. R.; Boyer, D. R.; Goldschmidt, L.; Rodriguez, J. A.; Cascio, D.; Chong, L.; Gonen, T.; Eisenberg, D. S. Atomic structures of low-complexity protein segments reveal kinked beta sheets that assemble networks. *Science* **2018**, *359*, 698–701.

(13) Luo, F.; Gui, X.; Zhou, H.; Gu, J.; Li, Y.; Liu, X.; Zhao, M.; Li, D.; Li, X.; Liu, C. Atomic structures of FUS LC domain segments reveal bases for reversible amyloid fibril formation. *Nature Structural & Molecular Biology* **2018**, *25*, 341–346.

(14) Meyer, D. E.; Chilkoti, A. Quantification of the Effects of Chain Length and Concentration on the Thermal Behavior of Elastin-like Polypeptides. *Biomacromolecules* **2004**, *5*, 846–851.

(15) Amiram, M.; Quiroz, F. G.; Callahan, D. J.; Chilkoti, A. A highly parallel method for synthesizing DNA repeats enables the discovery of 'smart' protein polymers. *Nat. Mater.* **2011**, *10*, 141–148.

(16) Quiroz, F. G.; Chilkoti, A. Sequence heuristics to encode phase behaviour in intrinsically disordered protein polymers. *Nature Materials* **2015**, *14*, 1164–1171.

(17) Varanko, A. K.; Su, J. C.; Chilkoti, A. Elastin-Like Polypeptides for Biomedical Applications. *Annu. Rev. Biomed. Eng.* **2020**, *22*, 343–369.

(18) Saha, S.; Banskota, S.; Roberts, S.; Kirmani, N.; Chilkoti, A. Engineering the architecture of elastin-like polypeptides: From unimers to hierarchical self-assembly. *Adv. Ther.* **2020**, *3*, 1900164.

(19) Sing, M. K.; Glassman, M. J.; Vronay-Ruggles, X. T.; Burghardt, W. R.; Olsen, B. D. Structure and rheology of dual-associative protein hydrogels under nonlinear shear flow. *Soft Matter* **2017**, *13*, 8511–8524.

(20) Glassman, M. J.; Olsen, B. D. Arrested Phase Separation of Elastin-like Polypeptide Solutions Yields Stiff, Thermoresponsive Gels. *Biomacromolecules* **2015**, *16*, 3762–3773.

(21) Glassman, M. J.; Avery, R. K.; Khademhosseini, A.; Olsen, B. D. Toughening of Thermoresponsive Arrested Networks of Elastin-Like Polypeptides To Engineer Cytocompatible Tissue Scaffolds. *Biomacromolecules* **2016**, *17*, 415–426.

(22) Advanced Photon Source 12-ID-B Home Page and Detector Configuration. Advanced Photon Source, Argonne National Laboratory, 2026; `https://www.aps.anl.gov/Sector-12/12-ID-B`, Accessed July 22, 2026.

(23) Xu, L.; Wang, Z.; Chen, J.; Chen, S.; Yang, W.; Ren, Y.; Zuo, X.; Zeng, J.; Wu, Q.; Sheng, H. Folded Network and Structural Transition in Molten Tin. *Nature Communications* **2022**, *13*, 126.

(24) Zheng, C. Y.; Yao, Y.; Deng, J.; Seifert, S.; Wong, A. M.; Lee, B.; Mirkin, C. A. Confined Growth of DNA-Assembled Superlattice Films. *ACS Nano* **2022**, *16*, 4813–4822.

(25) Teixeira, J. Small-Angle Scattering by Fractal Systems. *Journal of Applied Crystallography* **1988**, *21*, 781–785.

(26) Sunde, M.; Serpell, L. C.; Bartlam, M.; Fraser, P. E.; Pepys, M. B.; Blake, C. C. F. Common Core Structure of Amyloid Fibrils by Synchrotron X-Ray Diffraction. *Journal of Molecular Biology* **1997**, *273*, 729–739.